NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-83536/13
AMENDMENT 3
10 July 1999
SUPERSEDING
AMENDMENT 2
30 September 1996

PERFORMANCE SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, DPDT,
LOW LEVEL TO 10 AMPERES, MAGNETIC LATCH, HERMETICALLY SEALED,
ALL WELDED, TRANSIENT SUPPRESSED DC COIL

This amendment forms a part of MIL-PRF-83536/13, dated 27 March 1992, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

FIGURE 1, side view: Delete "SERIAL NUMBER SHALL BE MARKED ON TOP OF RELAYS".

PAGE 2

FIGURE 1, side view: Delete "SERIAL NUMBER SHALL BE MARKED ON TOP OF RELAYS".

PAGE 3

FIGURE 1, side view: Delete "SERIAL NUMBER SHALL BE MARKED ON TOP OF RELAYS".

PAGE 5

* NOTE 9, delete and substitute “9. Socket pin terminals shall provide the operational, environmental, and interface characteristics to provide a reliable interconnect to gold-plated contacts. Terminals, except the polarizing pin, shall be gold plated. One system for gold plating that may be used is ASTM B488, type 3, class 1.25 with a nickel underplate of 50 to 150 microinches thick. The gold plating system shall enable the product to meet the performance requirements of this specification and shall be approved by the qualifying activity.”

* NOTE 10, delete and substitute “10. Gasket shall provide a reliable seal between the relay and mating socket that will meet the environmental, operational, and interface requirements of the relay with the mating socket. The gasket shall have shore hardness 15 to 35, thickness .050 ±.005. Gasket material according to AMS 3332 has been considered acceptable.”

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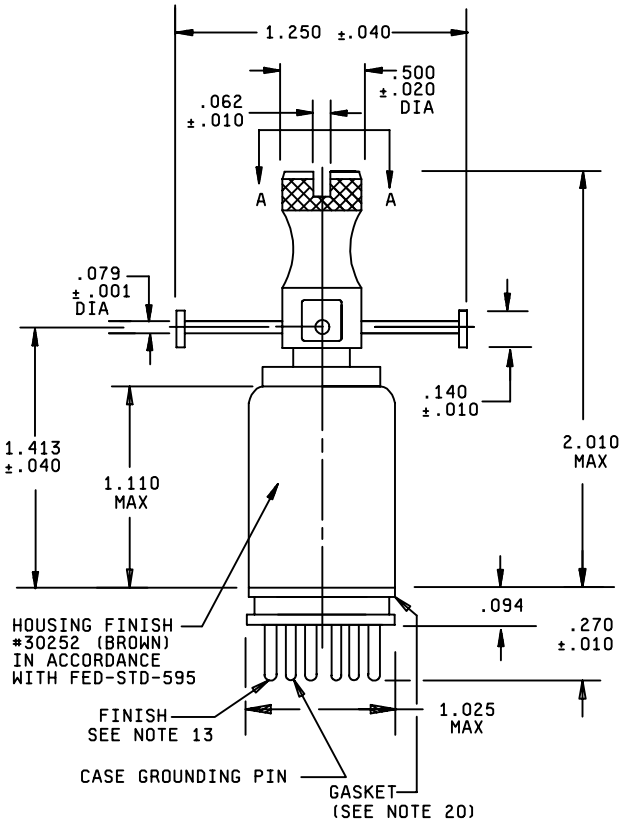
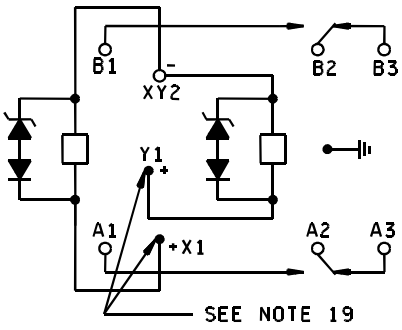
PAGE 6

* FIGURE 2, side view: Delete and substitute:



PAGE 7

FIGURE 2, add the following circuit diagram:



SEE NOTES 4, 5, AND 6

CIRCUIT DIAGRAM (TRACK MOUNT)
(OPTIONAL INTERNAL
DIODE CONFIGURATION)

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PAGE 8

* NOTE 13, delete and substitute “13. Socket pin terminals shall provide the operational, environmental, and interface characteristics to provide a reliable interconnect to gold-plated contacts. Terminals shall be gold plated. One system for gold plating that may be used is ASTM B488, type 3, class 1.25 with a nickel underplate of 50 to 150 microinches thick. The gold plating system shall enable the product to meet the performance requirements of this specification and shall be approved by the qualifying activity.”

* Add a NOTE 20, “20. Gasket shall provide a reliable seal between the relay and mating socket that will meet the environmental, operational, and interface requirements of the relay with the mating socket. The gasket shall have shore hardness 30 to 45, thickness .050 ±.005. Gasket material according to AMS 3332 has been considered acceptable.”

PAGE 9

Delete "Contact voltage drop or resistance:" and substitute "Contact voltage drop and resistance:".

Break bounce (normally open contacts) requirement: Delete in its entirety.

PAGE 10

Delete "Coil transient suppression: Applicable." and substitute "Coil transient suppression: Applicable (42 V dc maximum for track mount and 5 V dc maximum for all others)."

PAGE 11

TABLE III, delete and substitute:

"TABLE III. Dash numbers and characteristics. 1/

Dash number 2/		Mount	Coil data						
			At 25°C				Over temperature range	Maximum pickup voltage	
Solder hook	Socket pin		Coil voltage (V dc) 3/		Coil current (amperes) minimum	Specified latch/ reset voltage (V dc) 4/	Specified latch/ reset voltage (V dc) 4/	High temper-ature test	Contin-uous current test
			Rated	Max					
001	002	RVFM	6	7.3	0.22	3.2	4.5	5.0	5.7
003		HFM	6	7.3	0.22	3.2	4.5	5.0	5.7
004	005	RVFM	12	14.5	0.11	6.5	9.0	9.9	11.25
006		HFM	12	14.5	0.11	6.5	9.0	9.9	11.25
007	008	RVFM	28	29 5/	0.075	14.5	18.0	19.8	22.5
009		HFM	28	29 5/	0.075	14.5	18.0	19.8	22.5
	010	Track	28	29 5/	0.075	14.5	18.0	19.8	22.5
011	012	RVFM	48	50	0.040	24.0	36.0	38.0	42.0
013		HFM	48	50	0.040	24.0	36.0	38.0	42.0

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1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits are not recommended for subsequent use in low level applications.
2/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be L, 3.0; M, 1.0; P, 0.1; R, 0.01. Example: 001L, 002R.
3/ CAUTION: The use of any coil voltages less than the rated coil voltage will compromise the operation of the relay.
4/ Pickup voltage as shown is for test purposes only and is not to be used for design criteria.
5/ When maximum ambient temperature does not exceed +85°C, the maximum coil voltage shall be 32 V dc."

PAGE 12

Dielectric withstanding voltage, Track mount, Coil to coil/Coil to case, Sea level, After life: Delete "1250" and substitute "1050".

Dielectric withstanding voltage, Track mount, Altitude, At 300,000 feet: Delete "500" and substitute "N/A" (2 places).

Dielectric withstanding voltage, Track mount, All other points, Sea level, Initial: Delete "1000" and substitute "1500".

Delete "Maximum leakage current: 100 microamperes rms in accordance with MIL-R-83536." and substitute "Maximum leakage current for track mount only: 100 microamperes rms."

Vibration (random), first paragraph, line 2: Delete "test condition IJ" and substitute "test condition IG".

Vibration (random), second paragraph, line 2: Delete "15 minutes each plane".

Weight, delete in its entirety and substitute: "Weight: 0.90 pound (40.9 grams); 0.120 (54.5 grams) for track mount."

The margins of this amendment are marked with an asterisk to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based upon the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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(Project 5945-1059-13)

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